kakaohealthcare

Kakao Healthcare: Improving healthcare with a high-performance, safe, reliable data platform

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GOOGLE CLOUD RESULTS

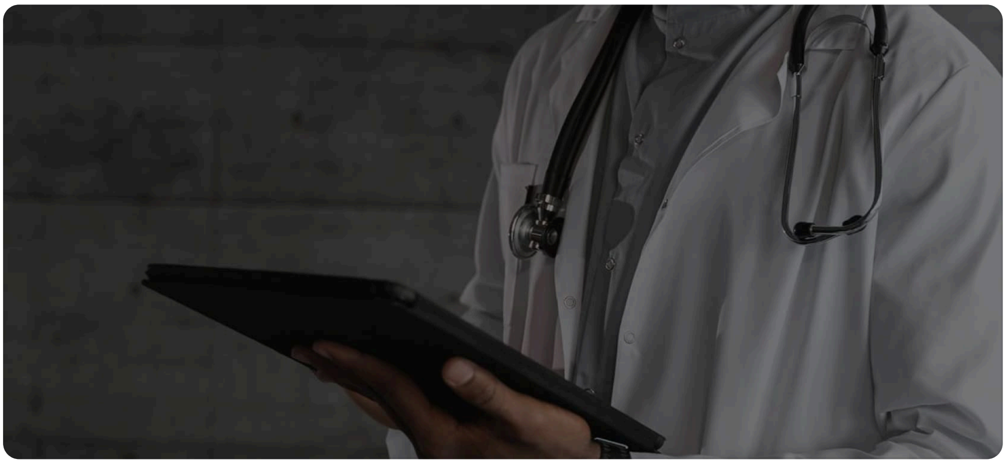
- Collaborative network enabled by AlloyDB empowers organizations for federative learning through data sharing
- Enables smooth data extraction, conversion, and loading with Dataproc to form a collaborative data network across hospitals
- Ability to seamlessly connect AI services, such as Gemini and Vertex AI, across its network
- Building a safe and collaborative medical data platform

With AlloyDB, Kakao Healthcare built a data platform that safely and systematically manages hospital medical

data and can be deployed and used across various medical settings in the cloud.

Kakao Healthcare aims to make people healthy through technology, using data and artificial intelligence (AI). With medical experience and a high understanding of hospital systems, the founders had an idea to improve classification and utilization systems for medical information and healthcare data to create better experiences.

From daily healthcare data to hospital treatment information, digital healthcare can overcome the limitations of traditional diagnosis and treatment-centered healthcare. Kakao Healthcare uses data to help support everything from smart solutions for hospital data platforms to diabetes management services for the general public and other IoT solutions. It focuses on the possibility of creating an environment where anyone can easily manage their health.



Turning passive data into an active business driver

“We were able to achieve similar performance through database tuning

Currently, most of the profits in domestic hospitals come from medical activities. As such, hospitals are looking toward making large investments in their medical services to grow revenue, but may not have the necessary know-how. South and hardware expansion, but we decided that it would not be appropriate to apply the same approach to all other hospitals. In the end, we decided that the cloud was most appropriate for all hospitals to quickly and easily build the same environment. This is when we considered Google Cloud, which specializes in data loading and analysis."

Hancheol Jekal
Head, Kakao Healthcare Data Platform Development

Korea has a robust medical information system because regulations require hospitals to store large amounts of health and clinical data. Because of this wealth of information, there's a growing demand to use that data to support treatment and research.

"Kakao Healthcare's data platform enhances the medical experience in hospitals by extracting new value from unused, high-quality healthcare data. That data also serves various organizations, including the government, pharmaceutical companies, medical AI companies, and medical services companies, working to improve treatments and health outcomes. We want to provide opportunities to create additional value in third-party environments that require data for growth," says Hancheol Jekal, Head of Kakao Healthcare Data Platform Development.

Hancheol adds that building a healthcare data platform began with increasing the value of data collected in medical settings and sharing it in a usable environment. The goal was to create a collaborative network so that participating hospitals could safely manage data while allowing it to be used in various environments where data is needed. To this end, Kakao Healthcare conducted a project to advance data solutions with a local university hospital.

The system was initially designed around an open source database based on PostgreSQL to be run on-premises within the hospital for versatility. However, because this hospital was already using an in-memory data analysis appliance, it already had an environment with very fast data processing speed. By applying the open source-based PostgreSQL as is, it was challenging to upgrade or establish a data platform at the level of the existing environment.

“We were able to achieve similar performance through database tuning and hardware expansion, but we decided that it would not be appropriate to apply the same approach to all other hospitals. In the end, we decided that the cloud was most appropriate for all hospitals to quickly and easily build the same environment. This is when we considered Google Cloud, which specializes in data loading and analysis.”

Kakao Healthcare signed a business agreement with Google to strengthen its data and healthcare business, and through this, established itself as a partner that thinks and creates a healthcare data platform business together, taking their relationship beyond just being a cloud provider and user.

Resolving technical limitations, including security, distribution, and performance

Performance and stability are particularly important for medical data. At the same time, strict security and privacy standards are required when medical and healthcare-related data are in the cloud. Hancheol reviewed several products together with Google Cloud, matching security standards and performance factors.

“We contemplated between [BigQuery] and [AlloyDB]. BigQuery was great, but AlloyDB was easy to apply to PostgreSQL, where the system was initially designed, and was effective in both technology

“Only when the number of participating hospitals and the size of data increases, will the effects of learning and analysis become meaningful. Google Cloud has provided an environment optimized for transfer and adoption and operating costs. I was also impressed by the fact that with the AlloyDB Columnar Engine, you could achieve performance at the level of a commercial in-memory database without hardware expansion or database tuning," says Hancheol.

this federated learning."

Hancheol Jekal
Head, Kakao Healthcare Data Platform Development

PostgreSQL is optimized for online transaction processing (OLTP), so if there are real-time analysis requirements, performance limitations would occur even if more central processing units (CPU) and memory resources were added. Meanwhile, AlloyDB can achieve higher performance with fewer resources, and has the advantage of being able to utilize the Columnar Engine without additional costs. Since AlloyDB is a fully-managed service, Kakao Healthcare saved time on maintenance and administrative tasks during the initial deployment.

Data extraction, conversion, and loading were done smoothly through Dataproc, making it advantageous to form a collaborative network where data from multiple hospitals were gathered to create greater value. Kakao Healthcare drew up an integrated data environment architecture using various data services from Google Cloud, and established a foundation that could be smoothly deployed and operated in actual medical settings.

Now, Kakao Healthcare has introduced its data platform to hospitals and records all clinical data. Through appropriate data preprocessing, all information that can identify patients is anonymized to fully protect privacy, and then loaded into AlloyDB in real time.

Through this data platform, patients' diseases, treatment details, drug prescriptions, rehabilitation, hospitalization period, and cure and survival rates are organically loaded and created in AlloyDB. The clinical results of the treatment process or prescription can be analyzed, and through AlloyDB, a database is provided to judge various hypotheses and results in the medical environment.

"I have high expectations for federated learning based on collaborative networks. Even clinical trials for drug development cannot be conducted within a single hospital to sample patients taking a specific drug. Only when the number of participating hospitals and the size of data increases, will the effects of learning and analysis become meaningful. Google Cloud has provided an environment optimized for this federated learning," says Hancheol.

With Google Cloud, Kakao Healthcare is now trying to collect hospital data. However, collaborative networks are not easily achievable because data is not only hospital assets, but also patients' private information. Therefore, data sharing becomes challenging, not only in terms of regulation, but also data ethics and technology.

When the Google Cloud-based data platform is operated at each hospital, a model tailored to the research topic is developed, and this can be applied to each data platform to learn and improve the model to find the desired results. AlloyDB has become a way to increase participation in healthcare organizations without the data being combined, shared or transmitted. To this end, Kakao Healthcare is currently supplying data platforms to major hospitals in Korea and building a collaborative network.

"The AlloyDB-based data platform is being deployed to tertiary level general hospitals, including university hospitals and large hospitals, starting in early 2024. AlloyDB delivers performance comparable to commercial in-memory databases at a reasonable cost. From the developer's perspective, we believe that data handling technologies are more easily accessible from a variety of angles, laying a solid foundation for upgrading solutions," says Hancheol.

He explains that AlloyDB creates an environment where data can be loaded, processed appropriately, and analyzed. Beyond data processing, Google Cloud carried a series of processes to effectively introduce analysis results internally to other researchers.

Increasing the value of data through large-scale analysis

"Combining AlloyDB, Dataproc, and AI

Technical support was also important. Upon signing the business agreement with Google Cloud, Kakao Healthcare and Google Cloud sought a common goal to services, we can create value for pharmaceutical and medical development companies that actually need extensive clinical data, participating in the development of a better medical environment while being profitable."

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effectively handle, distribute, and operate healthcare data.

"The technical support provided by Google Cloud was impressive during the process of introducing AlloyDB. From the process of designing the data platform, Google Cloud studied the architecture together and presented various methods to optimize the data environment, including AlloyDB. We suggested the most appropriate services to configure the data platform and provided close technical support," shares Hancheol.

Hancheol is considering ways to better handle data through AI, such as vision computing and natural language processing. Kakao Healthcare is already utilizing a variety of AI services, and with data being operated on Google Cloud, we have confirmed the possibility of easily and seamlessly connecting various AI services, such as Gemini and Vertex AI.

In the long run, Kakao Healthcare plans to bring the value of medical data beyond Korea, into the global medical market through its data platform. Currently, clinical data is concentrated in specific countries, but if data analysis is conducted in countries where patients actually receive treatment and take medicine, the accuracy of data can be greatly improved.

Hancheol also sees an opportunity for medical institutions starting other businesses based on datasets. The company is focused on the possibility of turning the value of data into a business model.

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Hancheol says that an environment in which hospitals and the medical market can continue to grow is needed. Medical IT is a sensitive and difficult field because it deals with life. For this reason, it has become natural to take a cautious and conservative approach to data management. However, the role of medical IT is to ensure that data creates great value for humanity.

Kakao Healthcare is also upgrading various other services to create a data environment that is accessible to benefit everyone, along with its responsibility to provide medical care. Together with Google Cloud data services, including AlloyDB, the company is moving toward that goal.

Kakao Healthcare seeks to solve the inconveniences experienced by users and partners of various healthcare services based on its core value of 'companion, friend, and secretary for everyone.' It provides digital healthcare services to contribute to the improvement of public health, targeting the global healthcare market through its mission to "keep people healthy through technology."

Industry: Healthcare & Life Sciences

Location: South Korea

Products: Google Cloud, AlloyDB, Dataproc